**2.**

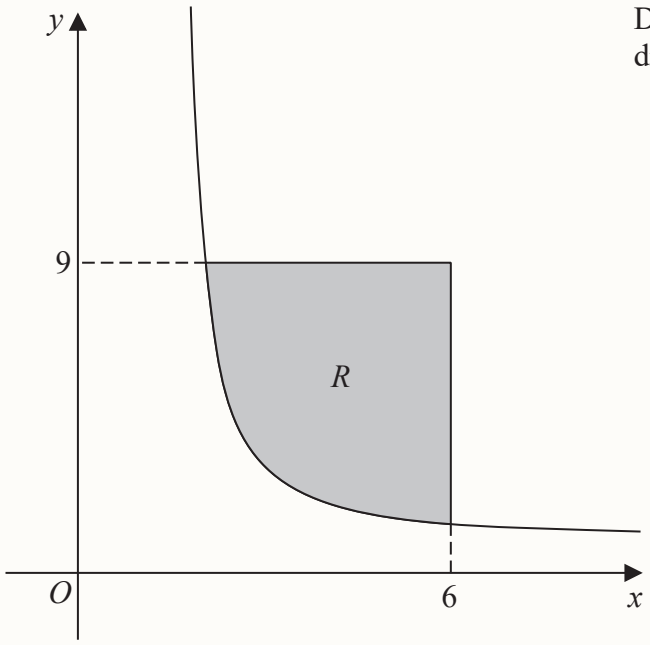



Diagram **NOT** drawn to scale

**Figure 1**

Figure 1 shows a sketch of part of the curve with equation

$$y = \frac{9}{(2x-3)^{1.25}} \qquad x > \frac{3}{2}$$

The finite region $R$, shown shaded in Figure 1, is bounded by the curve, the line with equation $y = 9$ and the line with equation $x = 6$

This region is rotated through $2\pi$ radians about the $x$-axis to form a solid of revolution.

Find, by algebraic integration, the exact volume of the solid generated.

**(7)**